INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

SLEEVING, TEXTILE, BRAIDED, SYNTHETIC POLYMER, -67 ºF (-55 ºC) TO +221 ºF (+105 ºC)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes the government acquisition requirements for one type of black, tubular, braided sleeving, in 14 sizes, which is used as protective sheathing over internal cable and wire bundles and harnesses in electronic equipment. This sleeving is made with a synthetic polymer, such as polyamide, and is suitable for use within a temperature range of -67 ºF (-55 ºC) to +221 ºF (+105 ºC).

2. CLASSIFICATION. The sleeving shall be classified by sizes listed in table I (see 7.4(b)).

TABLE I. Sleeving sizes.

Size code	Sleeving size (inches)	Size code	Sleeving size (inches)
1	1/8	8	3/4
2	3/16	9	7/8
3	1/4	10	1
4	5/16	11	1-1/4
5	3/8	12	1-1/2
6	1/2	13	1-3/4
7	5/8	14	2

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AMSC N/A FSC 5970

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3. SALIENT CHARACTERISTICS

3.1 Material. The sleeving shall be braided using cordage made from a synthetic polymer, preferably polyamide, which will permit the finished product to conform to the salient characteristics of this CID. Cordage shall be made by twisting together two or more ply yarns, ply yarns shall be made by twisting together two or more yarns, and yarns shall be made by twisting together two or more single continuous filament threads of the synthetic polymer. Joints or splices in the sleeving shall be limited to fused joints in yarn or cordage only. Joints shall not appear more often than once in 25 yards of sleeving, and not more often than 20 times in 600 yards.

3.2 Sleeving dimensions. Sleeving shall conform to the dimensions and construction details identified in table II.

TABLE II. Sleeving dimensions and construction characteristics.

Sleeving size	Inside diameter (inches nominal)	Number of ends	Number of carriers	Picks per inch	Wall thickness (inch)
1	1/8	24	24	20 ± 2	0.030 ± 0.015
2	3/16	32	32	18 ± 2	
3	1/4	44	44	20 ± 2	
4	5/16	44	44	18 ± 2	
5	3/8	48	48	18 ± 2	
6	1/2	64	64	16 ± 2	
7	5/8	128	64	12 ± 2	0.040 ± 0.015
8	3/4	128	64	11 ± 2	
9	7/8	192	64	10 ± 2	
10	1	216	72	10 ± 1	0.050 ± 0.015
11	1-1/4	216	72	9 ± 1	
12	1-1/2	288	72	8 ± 1	0.060 ± 0.015
13	1-3/4	288	72	7 ± 1	
14	2	360	72	6 ± 1	

3.2.1 Length. The length of the sleeving shall be specified in the acquisition order (see 7.4(c)).

3.3 Color. Unless otherwise specified, the sleeving shall be colored gloss black using a chrome fast dye.

3.4 Braid coverage. The overall braid coverage shall not be less than 85 percent and shall be verified by the procedure set forth in 3.4.1.

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3.4.1 Procedure. Sleeving specimens, each not less than 6 inches in length, shall be carefully cut and fitted over a smooth-surfaced mandrel that has hemispherical ends. The mandrel diameter shall correspond in size to the inside diameter of the sleeving within ±1/64 inch. The sleeving shall be fitted on the mandrel in a manner where it is not stretched (tightly fitted), loosely fitted, compressed, or gathered. Braid coverage shall then be determined by using the following formula:

$$K = 100(2F - F^2)$$

where K = percent of braid coverage
F = NP(0.0234)/sin a
N = number of ends per carrier
P = picks per inch of sleeving length
a = angle of braid with longitudinal axis of sleeving = arctan [2P(D + 0.0468)/C]
D = inside diameter of sleeving in inches
C = number of carriers

3.5 Tensile strength. The minimum breaking strength of jointless cords in the sleeving shall be not less than 25 pounds per square inch. The minimum breaking strength of fused cord joints shall not be less than 70 percent of the average breaking strength of the jointless cords.

3.6 Fungus resistance. The sleeving shall not support the growth of fungus.

3.7 Temperature. The sleeving shall be suitable for use as a protective sheath for internal electrical cable and wiring bundles and harnesses in spacecraft, aircraft, ship, ground vehicle, and shore-based electronic equipment throughout the temperature range of -67 ºF (-55 ºC) to +221 ºF (+105 ºC).

3.8 Workmanship. The sleeving shall be free of visible defects, dirt, grime, grease, and abraded or damaged surfaces.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

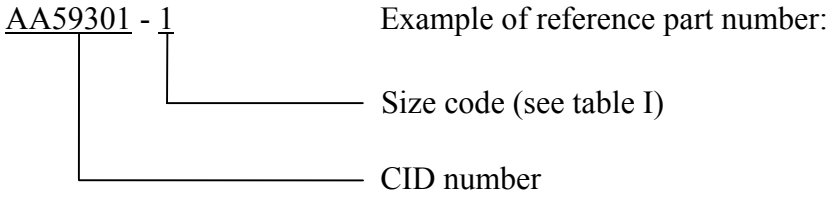
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6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as specified in the acquisition order (see 7.4(d)).

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 Sources of documents.

7.2.1 FAR. The FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of FAR documents may be obtained from http://www.arnet.gov/far/.

7.3 National stock number (NSN). The NSNs listed below are assigned to the listed product sizes (see table I). Other NSNs may also correspond with this document.

NSN	Size
5970-01-291-4842	2
5970-01-294-3468	3
5970-01-213-7413	4
5970-01-213-7412	5
5970-01-213-7410	6
5970-01-496-9102	6
5970-01-496-9406	8
5970-01-496-9259	11
5970-01-496-9256	12
5970-01-496-9254	13
5970-01-496-9252	14

7.4 Ordering data. The acquisition order should specify the following information:

a. CID document number, revision, and CID PIN.
b. Size (see 2).
c. Length of sleeving desired (see 3.2.1).
d. Preservation, packing, and marking requirements (see 6.1).

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7.5 Subject term (key word) listing.

cable
electronic
harnesses
polyamide
sheathing
tubular
wire
yarns

MILITARY INTERESTS:

Custodian:
Army - MI

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 7FLE

Preparing Activity:
DLA - GS2

(Project 5970-1266)

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